Endpoint crossings for the planar persistent walk

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Abstract

We study a persistent random walk on the square lattice, which is a lattice model of a run-and-tumble particle with four orientations. At each step the walk keeps its direction, turns left or right, or reverses, and turns and reversals are rare, with fixed ratios between their probabilities. The far corner $(N, 0)$, the lattice version of the ballistic front, is reached only by the path that starts east and never switches. The origin is reached by many paths that switch. For every even $N \geq 4$ the corner and the origin are equally likely at exactly one switching rate. At this crossing the expected number of switches is asymptotically a constant multiple of $\log N$, and we find the constant. When the walk reverses more often than it turns, the crossing agrees with that of the persistent walk on the line with $N/2$ steps, up to a negative power of N. When it turns more often, the crossing is set by a local limit theorem in two dimensions, and the logarithmic correction in the crossing equation is twice as large as on the line. On one curve of switching probabilities the planar walk is exactly two independent copies of the walk on the line, so the two crossings agree at every N. With turns only they agree up to a relative error of at most order $(\log N)/N$.

Keywords: persistent random walk, run-and-tumble particle, active matter, planar random motion, local limit theorem
MSC2020: 60G50, 60F05, 60J10, 82C41

1 Introduction

A persistent random walk on the square lattice $\mathbb{Z}^2$ moves one step east, north, west or south at a time. It starts in a uniform direction, and at each step it keeps its direction, turns left or right, or reverses, with given probabilities. In physics it is a lattice model of a run-and-tumble particle with four orientations. Such a particle is a standard model of an active, or self-propelled, particle such as a swimming bacterium. It moves in straight runs and picks a new orientation at random tumbles [7, 8]. Run-and-tumble particles with four orthogonal orientations are studied in [14, 15], and the continuous-time analogue of our walk is a planar random motion with orthogonal directions [16, 17, 18].

After N steps the corner $(N, 0)$ is reached only by the path that starts east and never switches, while the origin is reached by many paths that switch. We ask at which switching rate the two are equally likely. For an active particle the corner plays the role of the ballistic front. On the line the law of a run-and-tumble particle has atoms at the two ends of its support, carried by the particle that has not yet tumbled [9], and the four corners of our walk are the lattice version of these atoms.

The origin, in contrast, is reached only by paths that tumble and come back. The crossing is the tumbling rate at which the front site and the starting site are equally likely. In the continuum the corners carry atoms and the origin does not (it has a density in the plane, and with reversals also along the axes [18]), so the comparison of a corner with the origin is a lattice question.

On the line the same question compares an endpoint with the middle. The persistent random walk on the line [4, 5, 6, 2], which we call the line walk, steps right or left. Its first step is fair, and each later step repeats the previous one with probability p and switches with probability $q = 1-p$. An endpoint is reached by a single path, the one that never switches, so it has probability $\frac{1}{2}p^{N-1}$. When p is close to 1 this path wins, and when p is smaller the middle wins. For the line walk this question is part of Problem 131 of Kagey's Open Problem Collection [1], posed for a Galton board. It is answered at every N in [2]. The middle and an endpoint are equally likely when Nq, which is close to the expected number $(N-1)q$ of switches, is about $\log N$ (that is, the ratio of Nq to $\log N$ tends to 1). More precisely, the crossing value z_N of Nq satisfies $z_N + \frac{1}{2}\log z_N = \log N + \frac{1}{2}\log\frac{\pi}{8} + o(1)$.

1.1 Main results

Turns and reversals are rare. Fix $T > 0$. A turn to each side has probability rT/N and a reversal has probability sT/N, where $r, s \geq 0$ are fixed rates with $2r + s > 0$. So the walk switches about $(2r+s)T$ times on average, and this number is also its duration measured in units of the mean run time. Figure 1 shows the paths to the corner and to the origin. We show that for every even $N \geq 4$ there is exactly one value T_N^* of T at which the corner $(N, 0)$ and the origin are equally likely (Proposition 3.2). The corner is more likely below it and the origin above it, and as $N \to \infty$ through even integers

$$\frac{T_N^*}{\log N} \to \min\Big(\frac{2}{2r+s}, \frac{1}{s}\Big) \tag{1}$$

(Theorems 3.3 and 3.4, with $1/0 = \infty$). The two values in the minimum belong to two kinds of paths to the origin.

Why two values. The face selection rule of [3] for slowly switching Markov chains explains them. Call a set of directions a face, and call the vector of the numbers of steps in each direction the composition of a path. If $T = \tau \log N$, the most likely composition that uses exactly the directions of a face F has probability about $N^{-\tau\lambda_F - (|F|-1)}$, where λ_F is the rate at which the direction leaves F. The first term is the cost of never leaving F, and the second is the cost of sharing the mass of F among about $N^{|F|-1}$ compositions. The planar walk fits this picture. The corner is the face of the single direction east, with exit rate $2r + s$, so it has probability about $N^{-(2r+s)\tau}$. A path to the origin either stays on one axis, a face of two directions with exit rate $2r$, or uses all four directions, a face with exit rate 0. On an axis the origin is a single composition of that face, of probability about $N^{-(2r\tau+1)}$. With all four directions a single composition of the face has probability about N^{-3}, but about $N/2$ of them end at the origin, which gives N^{-2}. This suggests that the origin beats the corner when $(2r+s)\tau > \min(2r\tau + 1, 2)$, which is (1). It is only a heuristic, since the origin is not a single composition of the full face, and Section 3 proves the limit directly.

Second order. The mass at the origin splits exactly into the paths that stay on one axis and the paths that use all four directions (Proposition 3.2). When reversals dominate ($s > 2r$) the axis paths carry the origin mass at the crossing. Then $sT_N^* = z_{N/2} + O(N^{-\gamma})$ for some $\gamma > 0$, where $z_{N/2}$ is the crossing of the line walk with $N/2$ steps (Theorem 3.4). When turns dominate ($s < 2r$) the paths with all four directions carry it. Their mass is given by a local limit theorem in two

dimensions (Theorem 3.5), and the crossing then satisfies

$$(2r+s)\,T_N^* + \log T_N^* = 2\log N + \log\frac{\pi}{8(r+s)} + o(1)$$

(Corollary 3.6). Here $\log T_N^*$ has the coefficient 1, while the line walk has $\frac{1}{2}$, because the mass at the origin now comes from a local limit theorem in two dimensions.

Two copies of the line walk. There is one curve of switching probabilities on which everything is exact. If the walk turns to each side with probability $\rho(1-\rho)$ and reverses with probability ρ^2, its two diagonal coordinates are independent copies of the line walk with switching probability ρ. So on this curve the planar crossing is the crossing of the line walk at every N (Proposition 3.7). With turns only ($s = 0$) we show that $rT_N^* = z_N + O((\log N)^2/N)$ (Theorem 3.9).

1.2 Related work

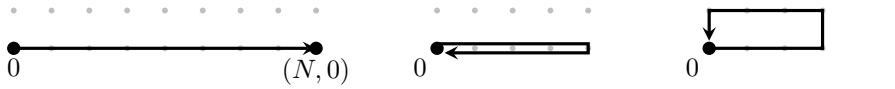
On the lattice the correlated random walk goes back to Gillis [10]. Chen and Renshaw [11] treat the Gillis–Domb–Fisher walk, and in [12] they show that the characteristic function of any correlated random walk satisfies a recurrence. So in principle the law of our planar walk is known. Marris and Giuggioli [13] give lattice propagators for walks with forward, backward and lateral steps. Lenci [22] proves recurrence for a large class of persistent walks on the square lattice.

In the continuum, Malakar *et al* [9] find the exact law of a run-and-tumble particle on the line and in an interval, and its first passage. In the plane, Santra, Basu and Sabhapandit [14] find the marginals of a run-and-tumble particle with four directions, and Angelani [15] treats switching among four orthogonal directions. With turns only, the continuous motion is a rotation of two independent telegraph processes (Cinque and Orsingher [16, Theorem 2.2], who credit Orsingher [17] for constant rates, and Smith, Le Doussal, Majumdar and Schehr [21]). Marchione and Orsingher [20] study orthogonal directions that change at Poisson times by clockwise or counterclockwise turns, write each coordinate as a linear combination of two independent telegraph processes, and obtain the exact distribution, including a singular part on the boundary. With reversals the two processes of the rotation are no longer independent [16, Section 3]. The curve of Proposition 3.7, on which the planar walk splits into two independent line walks, is the lattice analogue of these reductions. But on the lattice the independent case has reversals.

Orsingher and Marchione [19] study a continuous-time planar motion with orthogonal directions that turns clockwise, turns counterclockwise or reverses, each with its own probability. Our walk, in which the two turns are equally likely, is a discrete-time lattice analogue of this motion. They derive the singular distributions on the edges and diagonals of the square that bounds the motion, and its hydrodynamic limit. We did not find the planar crossing of the corner and the origin in the literature.

1.3 Plan of the paper

Section 2 fixes the notation for the line walk and recalls its crossing equation. Section 3 treats the planar walk for general rates. It splits the origin mass, proves the first-order answer, and locates the crossing when reversals dominate and when turns dominate. Section 3.1 (within Section 3) treats the product curve and the walk with turns only. Section 4 describes the computer checks and the open questions, and Appendix A has the longer proofs. From [2] we use the fact that the bin polynomials grow toward the middle, the uniform Bessel bound (recalled in (2)), the bound $z_N \le \log N$ and the expansion of z_N. The proofs do not use [3], which is the source of the heuristic above.



(a) the corner (b) an axis (c) all four directions

Figure 1: Paths of $N = 8$ steps. (a) The only path to the corner $(N, 0)$ never switches. A path to the origin either (b) stays on one axis and only reverses, or (c) turns and then uses all four directions.

2 Notation and the line walk

This section fixes the notation for the line walk and recalls its crossing equation from [2]. The planar walk itself is set up at the start of Section 3.

The line walk. For $a, b \geq 1$ let $S_{a,b}(u) = \sum_w u^{\mathrm{sw}(w)}$, the sum over the words w with a right steps and b left steps, where $\mathrm{sw}(w)$ is the number of switches of w. Each such word has probability $\frac{1}{2}p^{N-1}(q/p)^{\mathrm{sw}(w)}$ under the line walk with $N = a + b$ steps, and we say that it ends in bin a. So $S_{a,b}(q/p)$ is the probability of bin a divided by the probability $\frac{1}{2}p^{N-1}$ of an endpoint. For $N \geq 2$ let $S_N = S_{\lfloor N/2 \rfloor, \lceil N/2 \rceil}$, which belongs to a middle bin. It is a nonzero polynomial with nonnegative coefficients and no constant term, so $S_N(u) = 1$ has exactly one positive root u_N (the crossing value of the odds $u = q/p$ of a switch), and $p_N = 1/(1 + u_N)$ is the crossing of the middle and the endpoints. Put $L = \log N$, $z_N = N(1 - p_N)$ and $c_0 = \frac{1}{2}\log(\pi/8)$. By [2], z_N satisfies the crossing equation $z_N + \frac{1}{2}\log z_N = L + c_0 + 1/(8z_N) + R_N$ with an explicit remainder $R_N = O(1/L^2)$.

Conventions. The rates r and s are fixed. Unless we say otherwise, every constant, every implied constant in $O(\cdot)$ and every threshold for N depends only on them. Limits and $o(\cdot)$ refer to $N \to \infty$ through even integers. For a path of the planar walk we write n_j for its number of steps in the direction j, and we call the vector $n = (n_0, n_1, n_2, n_3)$ its composition. A set of directions is called a face, and the faces of one direction are called vertices. The exit rate of a face is the rate at which the direction of the walk leaves it (the principal Dirichlet eigenvalue in [3]). We need it only in the heuristics of the introduction and of Section 3, for a vertex, where it is $2r + s$, for an axis, where it is $2r$, and for all four directions, where it is 0.

3 The planar persistent walk

We now set up the planar persistent walk. It moves on $\mathbb{Z}^2$ in the directions east, north, west and south, numbered $0, 1, 2, 3$ modulo 4, with unit vectors $e_0 = (1, 0)$, $e_1 = (0, 1)$, $e_2 = -e_0$ and $e_3 = -e_1$. Let $r, s \geq 0$ with $\beta = 2r + s > 0$, and let $0 < h < 1/\beta$. The first direction is uniform. After that the walk keeps its direction with probability $p_0 = 1 - \beta h$, turns left or right with probability rh each, and reverses with probability sh. In the notation of [3] the direction is the chain with the rates $q_{j,j\pm1} = r$ and $q_{j,j+2} = s$. The position after N steps is $Y_N = (n_0 - n_2, n_1 - n_3)$. So Y_N is a linear function of the composition. Throughout, N is even, since otherwise the origin cannot be reached. We write $T = Nh$. Only the path that always goes east reaches the corner Ne_0, so

$$c_N = \mathbb{P}(Y_N = Ne_0) = \tfrac{1}{4}p_0^{N-1}.$$

We compare it with $o_N = \mathbb{P}(Y_N = 0)$. Figure 1 shows the three kinds of paths that matter.

We write $u = sh/p_0$, $v = rh/p_0$ and $w = \beta h/p_0 = 2v + u$, so $1 + w = 1/p_0$. These are the odds of a reversal, of a turn to a given side and of any switch, as in the notation $u = q/p$ of Section 2. We use S_N, u_N, p_N, $z_N = N(1 - p_N)$ and c_0 from Section 2. As N is even, $S_N(u)$ is the sum of

w^j over the words with $N/2$ letters of each of two kinds, where j is the number of switches. It has nonnegative coefficients and no constant term, so it increases strictly on $[0,\infty)$.

The answer at first order. The face selection of [3] predicts the answer, up to powers of $\log N$, as in the introduction. Let $h = \tau \log N/N$. The corner is the vertex $\{0\}$, which has exit rate β, so c_N is about $N^{-\beta\tau}$. A vertex, a face with three directions, or a face with two adjacent ones misses the opposite of one of its directions, so it cannot reach the origin. On the axis $\{0,2\}$, which has exit rate $2r$, the origin is the single composition $(N/2,0,N/2,0)$, of probability about $N^{-(2r\tau+1)}$. On the full face a single composition has probability about N^{-3}, but about $N/2$ compositions (a,b,a,b) end at the origin, and together they give about N^{-2}. So the origin should beat the corner when $\beta\tau > \min(2r\tau+1, 2)$, that is, when $\tau > \tau^* = \min(2/\beta, 1/s)$, with $1/0 = \infty$. At τ^* the axes win if $s > 2r$ and the full face wins if $s < 2r$. This is only a heuristic, since it applies the first-order face selection of [3] to a linear function of the composition, and no such rule is proved in general.

The rest of the section proves this and then locates the crossing. We split the origin mass into an axis part and a part with all four directions (Proposition 3.2) and prove the first-order answer (Theorem 3.3). Then we find the crossing up to $O(N^{-\gamma})$ when reversals dominate (Theorem 3.4) and up to $o(1)$ when turns dominate (Theorem 3.5 and Corollary 3.6). Section 3.1 treats two cases in which the walk is, exactly or nearly, two copies of the line walk.

Three levels. A path is given by the number of times it switches, the directions of its runs, and the lengths of its runs.

Lemma 3.1 (three levels)**.** *Write a sequence of directions as its number m of switches, its skeleton $Z = (Z_0,\dots,Z_m)$, which lists the directions of the runs so that $Z_t \neq Z_{t-1}$, and its run lengths $\ell = (\ell_0,\dots,\ell_m)$, which form a composition of N into $m+1$ positive parts. This is a bijection, and $Y_N = \sum_{t=0}^m \ell_t e_{Z_t}$. Under the walk,* (a) *$m$ has the law* $\mathrm{Bin}(N-1,\beta h)$. *Given m,* (b) *Z and ℓ are independent,* (c) *Z is the random walk on $\mathbb{Z}_4$ with uniform Z_0 whose independent increments are ±1 with probability r/β each and 2 with probability s/β, and* (d) *ℓ is uniform on the $\binom{N-1}{m}$ compositions.*

Proof. We write down the probability of one sequence and factor it. The map is a bijection because the runs of a sequence determine (m,Z,ℓ) and are determined by it. Put $\lambda(\pm1) = r$ and $\lambda(2) = s$. A sequence with data (m,Z,ℓ) starts in the direction Z_0 with probability $\frac14$, keeps its direction at $N-1-m$ steps, and makes the switch from Z_{t-1} to Z_t with probability $h\,\lambda(Z_t - Z_{t-1})$. So it has probability

$$\frac14 p_0^{N-1-m} \prod_{t=1}^m h\,\lambda(Z_t - Z_{t-1}) = \binom{N-1}{m}(\beta h)^m p_0^{N-1-m} \cdot \frac14 \prod_{t=1}^m \frac{\lambda(Z_t - Z_{t-1})}{\beta} \cdot \binom{N-1}{m}^{-1}.$$

The run lengths do not appear on the left, which is why they are uniform. On the right, the first factor depends only on m. For fixed m the middle factor is the probability of Z under the walk in (c), so it sums to 1 over the skeletons, and the last factor sums to 1 over the $\binom{N-1}{m}$ compositions. So the three factors are the laws in (a), (c) and (d), and the product form gives (b). □

The next proposition separates the skeletons that stay on one axis from those that use both.

Proposition 3.2 (exact decomposition and a unique crossing)**.** *Let $N \geq 2$ be even and $0 < h < 1/\beta$. Let $o_N^{(0)}$ and $o_N^{(4)}$ be the probabilities that $Y_N = 0$ and the walk never turns, and that $Y_N = 0$ and the walk uses all four directions. Then $o_N = o_N^{(0)} + o_N^{(4)}$,*

$$o_N^{(0)} = 2S_N(u)\,c_N, \qquad 0 \leq o_N^{(4)} \leq \frac{w^2}{2p_0}\bigl(1 - p_0^N\bigr).$$

Equivalently, $o_N/c_N = 2S_N(u) + E_N$ with $0 \le E_N \le 2w^2\big((1+w)^N - 1\big)$. Also, o_N/c_N is a polynomial in v and u with nonnegative coefficients and no constant term. If $s > 0$, or if $s = 0$ and $N \ge 4$, it increases strictly in h from 0 at $h = 0$ to ∞ as $h \uparrow 1/\beta$. So there is exactly one $h_N^* \in (0, 1/\beta)$ with $o_N = c_N$, and the corner is more likely than the origin exactly when $h < h_N^*$.

In words, the origin mass splits exactly into two parts. The axis part, measured against the corner, is twice the middle bin of the line walk with switching odds u, measured against its endpoint. The part from paths with all four directions is small when w is small. The ratio o_N/c_N grows with the switching rate, so the corner and the origin cross exactly once, at every even $N \ge 4$.

Proof. We sort the origin paths by whether they turn, and then we write o_N/c_N as a sum over paths. An origin path with no turn stays on the axis of its first step. An origin path that turns uses both axes, and since $n_0 = n_2$ and $n_1 = n_3$ at the origin, it uses all four directions. So every origin path is of exactly one of the two kinds. The origin paths on the axis $\{0, 2\}$ are the paths with $n_0 = n_2 = N/2$, and the path that always goes east has probability c_N. A path on this axis with j reversals has probability $\frac{1}{4}p_0^{N-1-j}(sh)^j = c_N u^j$. So by the definition of S_N these paths have total probability $S_N(u)c_N$, and the same holds on the axis $\{1, 3\}$. Hence $o_N^{(0)} = 2S_N(u)c_N$. (For a pair of states of a general chain this identity is proved in [3].) The bound on $o_N^{(4)}$ follows from Lemma 3.1 and a count of the run lengths that bring a skeleton back to the origin. We prove it in Appendix A. Dividing by c_N gives the form with E_N.

A path with j turns and k reversals has probability $\frac{1}{4}p_0^{N-1-j-k}(rh)^j(sh)^k = c_N v^j u^k$. So o_N/c_N is the sum of $v^j u^k$ over the origin paths, and $j + k \ge 1$ for each of them. Both u and v are nondecreasing in h, and each of them with a positive rate increases strictly to ∞ as $h \uparrow 1/\beta$. If $s > 0$, the path that goes east $N/2$ times and then west gives the term u. If $s = 0$ and $N \ge 4$, then $r > 0$, and a path whose four runs go east, north, west and south, with lengths $\ell_0 = \ell_2 \ge 1$ and $\ell_1 = \ell_3 \ge 1$, gives the term v^3. This proves the claims on monotonicity and the limits. For $s = 0$ and $N = 2$ the origin cannot be reached. □

The identity and the unique crossing for $s > 0$ are also checked in Lean 4 (Section 4). We write $T_N^* = Nh_N^*$ for the crossing and $u_N^* = sh_N^*/(1 - \beta h_N^*)$ for the odds of a reversal there. At the crossing $o_N = c_N$, so $2S_N(u_N^*) = o_N^{(0)}/o_N$ there. We call this number the zero-turn share of the origin mass.

We compare S_N with Bessel functions by the uniform Bessel bound of [2, Theorem 3.3(i)]. Put $\psi(z) = z(I_0(z) + I_1(z))$. In our notation that bound says that for every $N \ge 2$ (even or odd) and every $z > 0$

$$\Big(1 - \frac{z(z+2)}{2N}\Big)\psi(z) \le \frac{N}{2}\, S_N\Big(\frac{z}{N}\Big) \le \psi(z). \tag{2}$$

Since $I_0' = I_1$ and $(zI_1)' = zI_0$, we have $\psi' = \psi + I_0$, so $(\log\psi)' \ge 1$. We use this to turn a bound on $\psi(z)$ into a bound on z.

Theorem 3.3 (first order)**.** *Fix $r > 0$ and $s \ge 0$, put $\tau^* = \min(2/\beta, 1/s)$ with $1/0 = \infty$, and let $N \to \infty$ through even integers.*

(a) Fix $\tau > 0$ and put $h = \tau \log N/N$. If $\tau < \tau^*$, then $o_N/c_N \le N^{-\gamma}$ for some $\gamma > 0$ and all large N. If $\tau > \tau^*$, then $o_N/c_N \ge N^{\gamma}$ for some $\gamma > 0$ and all large N.

(b) The ratio $T_N^*/\log N$ tends to τ^*.

(c) If $s > 2r$, then $2S_N(u_N^*) \to 1$, so the origin mass at the crossing comes from the two axes, half from each. If $s < 2r$, then $2S_N(u_N^*) \le N^{-\gamma}$ for some $\gamma > 0$ and all large N, so it comes from the paths that use all four directions.

In words, the heuristic above gives the right first-order answer. The crossing sits at $T \approx \tau^* \log N$, and at the crossing the origin is reached through the axes when reversals dominate and through all four directions when turns dominate. (For $r = 0$ the walk never leaves the axis of its first step, and Theorem 3.4(c) gives the same limit $1/s$.)

Idea of the proof. The proof is in Appendix A. The upper bound in (a) follows from Proposition 3.2 and (2). For the lower bound when $s\tau > 1$ the axis part is enough. For the lower bound when $\beta\tau > 2$ we count balanced skeletons, those with as many runs east as west and as many north as south, with Lemma 3.1. They give $o_N^{(4)}$ of order at least $N^{-2}T^{-1/2}$. Part (b) follows from (a) and the monotonicity in Proposition 3.2, and part (c) from (b) and the bounds on the two parts.

Reversals dominate. Let $s > 2r$. Then the origin mass at the crossing comes from the axes, and on an axis the walk is the line walk. Up to an error $O(N^{-\gamma})$ the crossing is then the crossing of [2] for $N/2$ steps. We write $z_{N/2} = \frac{N}{2}(1 - p_{N/2})$ for it, and put

$$\mathcal{A}(\Lambda) = \Lambda - \tfrac{1}{2}\log\Lambda + c_0 + \frac{\frac{1}{4}\log\Lambda - \frac{1}{2}c_0 + \frac{1}{8}}{\Lambda},$$

so that the third-order expansion of z_N in [2, Corollary 3.7(c)] reads $z_N = \mathcal{A}(L) + O((\log L)^2/L^2)$.

Theorem 3.4 (reversals dominate)**.** *(a) Let $s > 0$. For all even $N \ge 2$ and $0 < h < 1/\beta$,*

$$\frac{o_N}{c_N} = 2S_N(u)\big(1 + \varepsilon_N(h)\big), \qquad 0 \le \varepsilon_N(h) \le \frac{w^2(1+w)^N}{S_N(u)}.$$

(b) Let $r \ge 0$ and $s > 2r$, let $N \to \infty$ through even integers, and let $0 < \gamma < 1 - 2r/s$. Then

$$\varepsilon_N(h_N^*) = O\big((\log N)^{3/2 - r/s} N^{-(s-2r)/s}\big), \qquad Nu_N^* = z_{N/2} + O(N^{-\gamma}).$$

(c) Under the hypotheses of (b), $sT_N^* = z_{N/2} + O(N^{-\gamma})$. Hence

$$sT_N^* = \mathcal{A}(L - \log 2) + O\Big(\frac{(\log L)^2}{L^2}\Big), \qquad sT_N^* - z_N = -\log 2 + \frac{\log 2}{2L} + O\Big(\frac{(\log L)^2}{L^2}\Big).$$

In words, when reversals dominate, the planar walk at its crossing behaves like the line walk with half as many steps, and sT_N^* is smaller than the crossing z_N of the line walk by about $\log 2$. Here is why. At the crossing $o_N \approx o_N^{(0)} = 2S_N(u)c_N$, so the crossing solves $2S_N(u) \approx 1$. By (2), applied with N and with $N/2$ steps, the equations $2S_N(z/N) = 1$ and $S_{N/2}(z/(N/2)) = 1$ both say that $\psi(z) \approx N/4$. So doubling S_N has the same effect as halving N.

Proof. (a) Since $s > 0$, $S_N(u) \ge 2u > 0$. Divide Proposition 3.2 by $2S_N(u)$. Then $\varepsilon_N(h) = E_N/(2S_N(u))$ and $E_N \le 2w^2(1+w)^N$.

(b) The proof has two steps. We first show that the four-direction part E_N is negligible at the crossing, so that the crossing nearly solves $2S_N(u) = 1$. Then we compare this equation with the equation $S_{N/2}(u) = 1$ of the crossing of the line walk for $N/2$ steps. Both equations become statements about ψ, and ψ grows so fast that nearly equal values force nearly equal arguments. Write $u^* = u_N^*$ and $z^* = Nu^*$, and let E^* be the value of E_N at $h = h_N^*$.

Step 1 (E^* is small). At the crossing $o_N/c_N = 1$, so Proposition 3.2 gives

$$2S_N(u^*) + E^* = 1.$$

Hence $S_N(u^*) \le \frac{1}{2} < 1 = S_N(u_N)$, and $u^* < u_N$ because S_N is increasing. Now $w = \beta u/s$ for every h, so the value of w at the crossing is below $w_N = \beta u_N/s$. The bound $E_N \le 2w^2(1+w)^N$ of Proposition 3.2 increases with w, so

$$E^* \le \eta_N := 2w_N^2(1+w_N)^N \le 2w_N^2 e^{Nw_N}.$$

We estimate η_N. The second-order expansion of z_N in [2, Corollary 3.7(b)] gives $Nu_N = z_N/p_N = L - \frac{1}{2}\log L + O(1)$. So $Nw_N = \frac{\beta}{s}(L - \frac{1}{2}\log L) + O(1)$, and

$$e^{Nw_N} = O\big(N^{\beta/s} L^{-\beta/(2s)}\big), \qquad w_N^2 = O\Big(\frac{L^2}{N^2}\Big).$$

In the product the exponent of N is $\beta/s - 2 = -(s-2r)/s$, and the exponent of L is $2 - \beta/(2s) = \frac{3}{2} - \frac{r}{s}$. Hence

$$\eta_N = O\big(L^{3/2 - r/s} N^{-(s-2r)/s}\big),$$

which tends to 0 because $s > 2r$. Since $2S_N(u^*) = 1 - E^*$, part (a) gives $\varepsilon_N(h_N^*) = E^*/(1-E^*) \le \eta_N/(1-\eta_N)$. This is the first claim.

Step 2 (comparison with $N/2$ steps). Let $z' = \frac{N}{2} u_{N/2}$, so that $S_{N/2}(z'/(N/2)) = 1$. We apply (2) twice. With $N/2$ steps at z' it reads $(1 - \delta_N')\psi(z') \le \frac{N}{4} \le \psi(z')$, where $\delta_N' = z'(z'+2)/N$. With N steps at z^* it reads $(1-\delta_N)\psi(z^*) \le \frac{N}{2} S_N(u^*) \le \psi(z^*)$, where $\delta_N = z^*(z^*+2)/(2N)$, and Step 1 gives $\frac{1}{4}(1-\eta_N) \le \frac{1}{2} S_N(u^*) \le \frac{1}{4}$. Together,

$$\frac{N}{4} \le \psi(z') \le \frac{N}{4} \cdot \frac{1}{1-\delta_N'}, \qquad \frac{N}{4}(1-\eta_N) \le \psi(z^*) \le \frac{N}{4} \cdot \frac{1}{1-\delta_N}.$$

So ψ is close to $N/4$ at both points. Both z' and $z^* < Nu_N$ are $O(L)$, so δ_N and δ_N' are $O(L^2/N)$, and taking logarithms gives $|\log\psi(z^*) - \log\psi(z')| = O(\eta_N + L^2/N)$. Since $(\log\psi)' \ge 1$, two points are never farther apart than their values of $\log\psi$. So

$$|z^* - z'| \le |\log\psi(z^*) - \log\psi(z')| = O\Big(\eta_N + \frac{L^2}{N}\Big).$$

Finally $z_{N/2} = \frac{N}{2}(1 - p_{N/2}) = z'/(1+u_{N/2})$ and $u_{N/2} = O(L/N)$, so $z_{N/2} = z' + O(L^2/N)$. Hence $z^* - z_{N/2} = O(\eta_N + L^2/N)$. By the bound on η_N in Step 1 this is $O(N^{-\gamma})$, because $\gamma < 1 - 2r/s$.

(c) We pass from Nu_N^* to sT_N^* and then insert the expansion of z_N. Since $u = sh/(1-\beta h)$, we have $h_N^* = u^*/(s + \beta u^*)$. So $sT_N^* = z^*/(1 + \beta u^*/s) = z^* + O(L^2/N)$, as $u^* < u_N = O(L/N)$. With (b) this gives $sT_N^* = z_{N/2} + O(N^{-\gamma})$. The third-order expansion of z_N, for $N/2$ steps and for N steps, gives $z_{N/2} = \mathcal{A}(L - \log 2) + O((\log L)^2/L^2)$ and $z_N = \mathcal{A}(L) + O((\log L)^2/L^2)$, since $\log(N/2) = L - \log 2$. This is the first display. For the second we subtract. The last term of $\mathcal{A}$ has derivative $O(\log L/L^2)$ near L, so

$$\mathcal{A}(L - \log 2) - \mathcal{A}(L) = -\log 2 - \tfrac{1}{2}\log\Big(1 - \frac{\log 2}{L}\Big) + O\Big(\frac{\log L}{L^2}\Big) = -\log 2 + \frac{\log 2}{2L} + O\Big(\frac{\log L}{L^2}\Big). \quad \square$$

For $(r,s) = (0.1, 1)$ a numerical computation of the exact law gives the difference $sT_N^* - z_N$ as -0.6317 at $N = 640$ and -0.6493 at $N = 10240$, while the first two terms in (c) give -0.6395 and -0.6556 (Figure 2(a)).

Turns dominate. Let $s < 2r$. Now the origin mass at the crossing is carried by the paths with all four directions, and Proposition 3.2 only bounds $o_N^{(4)}$ from above. We need its size, which is a local limit theorem in two dimensions.

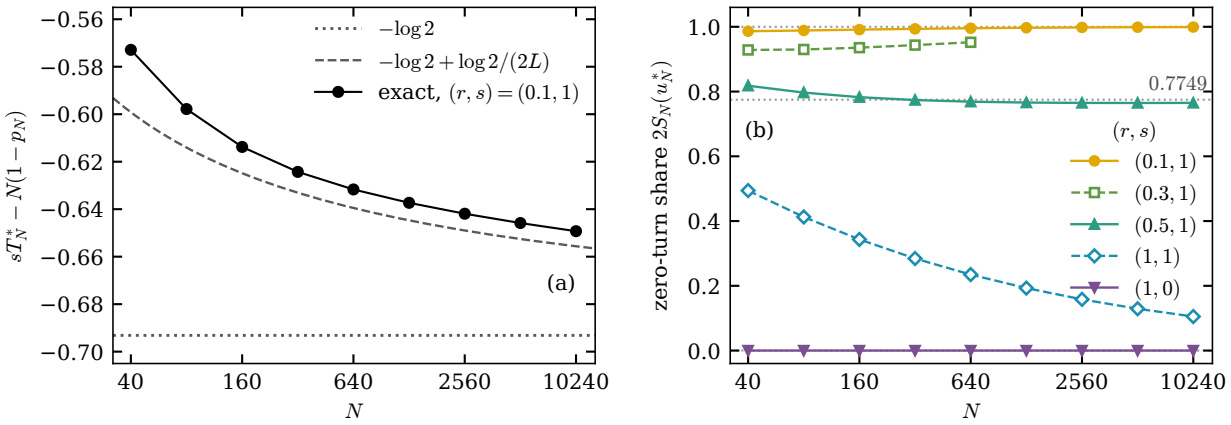


Figure 2: (a) The exact crossing of the planar walk with $(r, s) = (0.1, 1)$, written as $sT_N^* - z_N$ with $z_N = N(1 - p_N)$, for $N = 40 \cdot 2^k$ with $0 \le k \le 8$. The dashed curve is $-\log 2 + \log 2/(2L)$ from Theorem 3.4(c), and the dotted line is its limit $-\log 2$. (b) The zero-turn share $2S_N(u_N^*)$ of the origin mass at the crossing. It tends to 1 for $s > 2r$ and to 0 for $s < 2r$ (Theorem 3.3(c)), and to $2\sqrt{2}/(\sqrt{2} + \sqrt{5}) = 0.7749$ at the switch $s = 2r$ (Corollary 3.6(b)). For $(r, s) = (1, 0)$ there are no reversals and the share is 0 for every N. This is the walk with turns only of Theorem 3.9. The pair $(0.3, 1)$ was computed up to $N = 640$.

Theorem 3.5 (four-direction local limit)**.** *Fix $r > 0$ and $s \ge 0$. Let $N \to \infty$ through even integers, let $T = T_N \to \infty$ with $T_N \le N^{1/8}$, and put $h = T/N$. Then*

$$N^2 o_N^{(4)} = \frac{2(r+s)}{\pi}\, T\,\big(1 + o(1)\big).$$

In words, $o_N^{(4)}$ is asymptotically twice the density at 0 of a centered Gaussian vector with covariance $\frac{N^2}{2(r+s)T} I_2$ (the factor 2 appears because Y_N lies on a sublattice of index 2). Here is a heuristic for the variance, which is not the proof. The functions $\cos(\pi j/2)$ and $\sin(\pi j/2)$ of the direction j are the two coordinates of a step. They are eigenvectors, with eigenvalue $-2(r+s)$, of the generator Q of the direction chain (namely $Q_{j,j\pm1} = r$, $Q_{j,j+2} = s$ and $Q_{jj} = -\beta$). A coordinate of a step is ± 1 half of the time and 0 otherwise. Since the first direction is uniform, the directions are stationary. So the same coordinate of two steps that are k steps apart has covariance $\frac{1}{2}(1 - 2(r+s)h)^k$, which is about $\frac{1}{2}e^{-2(r+s)Tk/N}$. Summing over all pairs of steps gives the variance $N \cdot \frac{1}{2} \cdot \frac{2N}{2(r+s)T} = N^2/(2(r+s)T)$ for a coordinate of Y_N, up to a factor $1 + o(1)$. Our proof needs $T \le N^{1/8}$. Below we use the theorem only for T of order $\log N$.

Idea of the proof. The proof is in Appendix A. By Lemma 3.1 we may fix the skeleton and let the run lengths be uniform. The total lengths in the four directions are then close to N times a Dirichlet vector, and the chance that east cancels west and north cancels south is close to $N^{-2}g$, with an explicit function g of the numbers of runs in the four directions. Divided by m, this g is a Gaussian function of $W_m/\sqrt{m}$, where W_m is the sum of the unit vectors of the runs, and a martingale central limit theorem for W_m gives the constant.

At the crossing $o_N^{(4)} = c_N(1 - 2S_N(u_N^*))$, and with Theorem 3.5 this becomes an equation for T_N^*.

Corollary 3.6 (turns dominate, and the switch)**.** *Let $r > 0$ and let $N \to \infty$ through even integers.*

(a) If $0 \le s < 2r$, then

$$\beta T_N^* + \log T_N^* = 2L + \log \frac{\pi}{8(r+s)} + o(1),$$

or equivalently $\beta T_N^* = 2z_N + \log \frac{\beta}{2(r+s)} + o(1)$.

(b) If $s = 2r$, then $2S_N(u_N^*) \to \frac{2\sqrt{2}}{\sqrt{2}+\sqrt{5}} = 0.774852\ldots$ and

$$\beta T_N^* + \log T_N^* = 2L + \log \frac{\pi}{8(r+s)} + \log \frac{\sqrt{5}-\sqrt{2}}{\sqrt{5}+\sqrt{2}} + o(1).$$

In words, when turns dominate the crossing equation has $\log T$ with the coefficient 1, where the line walk has $\frac{1}{2}$. The reason is that the mass at the origin now comes from a local limit theorem in two dimensions. In one dimension the density at the center is of order $T^{1/2}/N$, and in two dimensions it is of order T/N^2. At the switch $s = 2r$ neither kind of path wins, and the two shares of the origin mass tend to explicit constants.

Proof. Let $0 \le s \le 2r$. At the crossing Proposition 3.2 gives $o_N^{(4)} = c_N(1 - 2S_N(u_N^*))$. We take logarithms in this identity and insert Theorem 3.5 on the left.

Since $\tau^* = 2/\beta$, Theorem 3.3(b) gives $T_N^*/L \to 2/\beta$. So $T_N^* \to \infty$ and $T_N^* \le N^{1/8}$ for large N, and Theorem 3.5 applies with $T_N = T_N^*$. It gives

$$\log o_N^{(4)} = -2L + \log T_N^* + \log \frac{2(r+s)}{\pi} + o(1).$$

On the right, $\log c_N = -\log 4 + (N-1)\log(1 - \beta h_N^*) = -\log 4 - \beta T_N^* + O(L^2/N)$. Both sides of the identity are positive, so we may equate their logarithms. This gives

$$\beta T_N^* + \log T_N^* = 2L + \log \frac{\pi}{8(r+s)} + \log\bigl(1 - 2S_N(u_N^*)\bigr) + o(1). \tag{3}$$

(a) By Theorem 3.3(c), $2S_N(u_N^*) \to 0$, so the last logarithm in (3) is $o(1)$. For the second form note that $\log T_N^* = \log(2L/\beta) + o(1)$, and that $2z_N = 2L - \log L + \log \frac{\pi}{8} + o(1)$ by the crossing equation of [2] recalled in Section 2.

(b) Now $\beta = 2s$ and $2(r+s) = 3s$. Here the axis part and the four-direction part of the origin mass have the same order at the crossing. We write both in terms of the one quantity $\varrho_N = Ne^{-sT_N^*}/\sqrt{T_N^*}$ and solve for it.

Put $z^* = Nu_N^*$, so that $z^* = sT_N^* + O(L^2/N)$ and $z^* \to \infty$. As in the proof of Theorem 3.3, (2) and $\psi(z) \sim \sqrt{2z/\pi}\, e^z$ give the axis part,

$$2S_N(u_N^*) = \frac{4}{N}\sqrt{\frac{2z^*}{\pi}}\, e^{z^*}(1 + o(1)) = \frac{a}{\varrho_N}(1 + o(1)), \qquad a = 8\sqrt{\frac{s}{2\pi}}.$$

Theorem 3.5 with $c_N^{-1} = 4e^{2sT_N^*}(1 + o(1))$ gives the four-direction part,

$$\frac{o_N^{(4)}}{c_N} = \frac{3s}{\pi} \cdot \frac{T_N^*}{N^2} \cdot 4e^{2sT_N^*}(1 + o(1)) = \frac{b}{\varrho_N^2}(1 + o(1)), \qquad b = \frac{12s}{\pi}.$$

The two parts are positive and add up to 1. So ϱ_N is bounded and bounded away from 0 (otherwise the sum would tend to ∞ or to 0 along a subsequence), and each of its limit points $\tilde{\varrho}$ solves $\tilde{\varrho}^2 - a\tilde{\varrho} - b = 0$. As $a^2 = 32s/\pi$ and $a^2 + 4b = 80s/\pi$, the only positive root is $\tilde{\varrho} = 2\sqrt{s/\pi}\,(\sqrt{2} + \sqrt{5})$. So ϱ_N converges to it. Hence $2S_N(u_N^*) \to a/\tilde{\varrho} = 2\sqrt{2}/(\sqrt{2} + \sqrt{5})$, so $1 - 2S_N(u_N^*) \to (\sqrt{5} - \sqrt{2})/(\sqrt{5} + \sqrt{2})$, and (3) gives the display. □

Dividing (a) or (b) by β gives $T_N^* = \frac{2}{\beta}[L - \frac{1}{2}\log L + O(1)]$ for $s \le 2r$. With Theorem 3.4(c) for $s > 2r$ this shows that $T_N^* = \tau^*[L - \frac{1}{2}\log L + O(1)]$ for all rates with $\beta > 0$. So the coefficient $\frac{1}{2}$ of $\log\log N$ in the crossing equation of [3] appears here too, even when the origin mass comes from a line of compositions and not from a single one. The approach to these limits is slow. In numerical computations of the exact law at $N = 10240$ the zero-turn share is 0.7650 for $(r, s) = (0.5, 1)$, where its limit is 0.7749, and still 0.1051 for $(1, 1)$, where its limit is 0 (Figure 2(b)). The errors in Theorem 3.5 and Corollary 3.6 are only $o(1)$, and we do not know the next term.

3.1 The product curve and turns only

The results so far are asymptotic when turns matter. There is one curve in the plane of the two switching probabilities on which everything is exact, because there the planar walk is two independent copies of the line walk. Rotate the lattice by 45°, that is, write $Y_N = (Y^{(1)}, Y^{(2)})$ and use $U = Y^{(1)} + Y^{(2)}$ and $V = Y^{(1)} - Y^{(2)}$. Each step changes U by ± 1 and V by ± 1, so a path of the planar walk is a pair of words in the letters $\pm$, namely the sign word of U and the sign word of V. A left turn changes the sign of one of the two and a right turn changes the other, while a reversal changes both.

Proposition 3.7 (the product curve)**.** *Let $N \ge 1$ and $0 < \rho < 1$. Suppose the planar walk keeps its direction with probability $(1-\rho)^2$, turns to each side with probability $\rho(1-\rho)$ and reverses with probability ρ^2 (so $v = x$ and $u = x^2$ with $x = \rho/(1-\rho)$). Then the sign words of U and V are independent copies of the walk of [2] with switching probability ρ. Hence, with P_N^{A} the law of the bin of the line walk with N steps and $q = \rho$,*

$$\mathbb{P}\big(Y_N = (y_1, y_2)\big) = P_N^{\mathrm{A}}\Big(\frac{N + y_1 + y_2}{2}\Big)\, P_N^{\mathrm{A}}\Big(\frac{N + y_1 - y_2}{2}\Big)$$

for all integers y_1, y_2 (the right side is 0 unless both arguments are integers in $[0, N]$). In particular, for even N,

$$\frac{o_N}{c_N} = S_N(x)^2,$$

and the corner and the origin are equally likely exactly when $\rho = 1 - p_N$.

In words, on this curve the two diagonal coordinates of the planar walk do not see each other. So the planar crossing is the crossing of the line walk, exactly and at every N, and every bound on p_N in [2] holds for it.

Proof. The steps e_0, e_1, e_2, e_3 change (U, V) by $(+,+)$, $(+,-)$, $(-,-)$ and $(-,+)$. So a direction is the same thing as a pair of signs, and a uniform first direction is a pair of independent fair signs. From any direction one turn changes only the sign of U, the other turn changes only the sign of V, and a reversal changes both. Under the hypothesis, at each step neither sign changes with probability $(1-\rho)^2$, only the first with probability $\rho(1-\rho)$, only the second with probability $\rho(1-\rho)$, and both with probability ρ^2, whatever the past. These are the probabilities of two independent events of probability ρ each. So the two sign words are independent, and each is a fair first letter followed by switches of probability ρ. The word of U has $(N+U)/2$ letters $+$, and likewise for V, which gives the product formula. The corner Ne_0 has $U = V = N$ and the origin has $U = V = 0$, so $o_N/c_N = (P_N^{\mathrm{A}}(N/2)/P_N^{\mathrm{A}}(N))^2 = S_N(x)^2$, and this equals 1 exactly when $x = u_N$, that is, when $\rho = 1 - p_N$. □

On the curve $\beta h = 1-(1-\rho)^2 = \rho(2-\rho)$, so the crossing is at $\beta T_N^* = 2z_N - z_N^2/N$. Corollary 3.6(a) is stated for fixed rates and does not cover this case, but the two agree in form. On the curve the probability of a reversal divided by that of a turn to a given side is $\rho/(1-\rho)$. At the crossing $\rho = 1-p_N \to 0$, so this ratio tends to 0. If s/r is read as this ratio, the constant $\log\frac{\beta}{2(r+s)}$ of Corollary 3.6(a) tends to 0 as well.

In continuous time the walk with turns only is a rotation of two independent telegraph processes (Orsingher [17], Cinque and Orsingher [16, Theorem 2.2], and Smith, Le Doussal, Majumdar and Schehr [21], see also [15, Eq. (36)]). With reversals the two processes are dependent [16, Section 3]. On the lattice two independent switches happen at the same step with probability ρ^2, which is why the curve has reversals. At $\rho = \frac{1}{2}$ this is the classical fact that the simple random walk in the plane is a rotation of two independent walks on the line. We did not find the lattice statement for other ρ.

Off the curve the two sign words are dependent. But o_N/c_N is increasing in each of the odds v and u, so the curve still gives bounds at every N.

Corollary 3.8 (comparison with the product curve)**.** *Let $N \geq 2$ be even and $0 < h < 1/\beta$. Then*

$$S_N\big(\min(v,\sqrt{u})\big)^2 \leq \frac{o_N}{c_N} \leq S_N\big(\max(v,\sqrt{u})\big)^2.$$

So at the crossing u_N lies between v and $\sqrt{u}$.

In words, the odds u_N at the crossing of the line walk lie between the odds of a turn to a given side and the square root of the odds of a reversal, both taken at the planar crossing.

Proof. By Proposition 3.2, o_N/c_N is a polynomial $\Phi_N(v,u)$ with nonnegative coefficients, so it is nondecreasing in v and in u. By Proposition 3.7, $\Phi_N(x,x^2) = S_N(x)^2$ for every $x > 0$. With $x = \max(v,\sqrt{u})$ we have $v \leq x$ and $u \leq x^2$, so $\Phi_N(v,u) \leq \Phi_N(x,x^2)$. With $x = \min(v,\sqrt{u})$ we get the lower bound in the same way (if $x = 0$ it is trivial). At the crossing $\Phi_N = 1 = S_N(u_N)^2$, and S_N is increasing. □

With turns only ($s = 0$) the sign words of U and V are two copies of the line walk that never switch at the same step. The upper bound of Corollary 3.8 then reads $o_N/c_N \leq S_N(v)^2$. The next theorem shows that forbidding common switches changes the crossing very little.

Theorem 3.9 (turns only)**.** *Let $s = 0 < r$ and let $N \geq 4$ be even. Call a word of length N in the letters $\pm$ balanced if it has $N/2$ letters of each sign, and let $\Delta_N(x)$ be the sum of $x^{j+j'}$ over the pairs of balanced words, with j and j' switches, that switch at a common step.*

(a) The identity $o_N/c_N = S_N(v)^2 - \Delta_N(v)$ holds and $\Delta_N \geq 0$. So the value v_N^* of v at the crossing satisfies $v_N^* \geq u_N$.

(b) If $z = Nx$ satisfies $z(z+2) \leq N$, then $\Delta_N(x) \leq \big(2x + 8xS_N(x)\big)(2z+2)\,S_N(x)$.

(c) As $N \to \infty$,

$$Nv_N^* = Nu_N + O\Big(\frac{L^2}{N}\Big) \qquad \text{and} \qquad rT_N^* = z_N + O\Big(\frac{L^2}{N}\Big).$$

In words, with turns only the planar crossing is the crossing of the line walk up to $O(L^2/N)$, and measured by the odds of a turn it is never earlier. Note that v_N^* and $rT_N^* = Nv_N^*/(1+2v_N^*)$ do not depend on r. At $N = 10240$ we find $v_N^*/u_N - 1 = 4.0\times10^{-4}$ and $rT_N^* - z_N = -0.0027$,

while $L^2/N = 0.0083$. (The second number is negative although $v_N^* \geq u_N$, because the odds v correspond to $rT = Nv/(1+2v)$ and the odds u_N to $z_N = Nu_N/(1+u_N)$.)

Idea of the proof. Part (a) is a count of paths. For (b) we use a union bound over the step of a common switch. Near the crossing, where $S_N(x)$ is close to one, a balanced word switches at a given step with probability of order x (Lemma A.4), and it has of order z switches in all. So a second balanced word shares a switch with it with probability of order $xz = z^2/N$. Part (c) follows from (a), (b) and the uniform Bessel bound (2). The proofs of (b) and (c) are in Appendix A.

Proof of (a). A path with turns only is a pair of words that never switch at the same step, and it has probability $c_N v^{j+j'}$ if the words have j and j' switches. It ends at the origin exactly when both words are balanced. Since $S_N(v)$ is the sum of v^j over the balanced words, $S_N(v)^2$ is the sum of $v^{j+j'}$ over all pairs of balanced words, and removing the pairs with a common switch leaves o_N/c_N. So $S_N(v_N^*)^2 \geq 1 = S_N(u_N)^2$, and S_N is increasing. □

4 Verification and open questions

The theorems, propositions, lemmas and corollaries of this paper are proved in the text or in the appendix. We quote results of [2] where we say so (for example the uniform Bessel bound (2)). No proof rests on a floating-point computation. The computed numbers in the paper, namely the planar crossings and zero-turn shares in Section 3 and Figure 2, are computations of the law at a finite N. Each crossing is found by one method and, for $N \leq 2560$, checked by a second, independent one. For larger N it rests on one method. At each computed crossing the zero-turn share is evaluated in two independent ways. These numbers illustrate the theorems, and no statement is deduced from them.

The script `verify_planar.py` computes o_N by the three levels of Lemma 3.1, by Fourier inversion, by a dynamic program and by a sum over all 4^N words. It compares the first three for even $N \leq 10$ and all four for $N \leq 8$. It finds the crossings by the first method and checks them by the second for $N \leq 2560$. It also tests the identity of Theorem 3.9(a) in exact rational arithmetic, for even $N \leq 30$ and four values of x. The form it tests is that o_N/c_N is the sum of $x^{j+j'}$ over the pairs of balanced words with no common switch. It also checks some bounds at finite N that are not statements of this paper.

The script `verify_planar_product.py` checks the product formula of Proposition 3.7 in exact rational arithmetic for $N = 4, 6, 8$, with one value of ρ each. Its other checks are in floating point and only on grids. The first is the inequality $F(y/2) \leq 2F(y)$ of Appendix A, at 4000 values of y between 10^{-3} and 10^4. The second is the inequality $xS_N'(x) \leq (2z+2)S_N(x)$ of Lemma A.4, at $N = 16$, 64 and 256 and 30 values of z with $z(z+2) \leq N$, where the derivative is a central difference. The third takes $N = 8, 16, 32, 64, 96$ and $x = \frac{1}{2}u_N, u_N, \frac{3}{2}u_N$. It checks that $\Delta_N(x) \geq 0$, and it checks the bound of Theorem 3.9(b) at those points where the hypothesis $z(z+2) \leq N$ holds. This script computes Δ_N as $S_N^2 - o_N/c_N$, so it does not test the identity of Theorem 3.9(a) a second time. A check that fails ends either script with an error. The script also prints $N(v_N^* - u_N)$ against L^2/N and does not test it, because Theorem 3.9(c) gives no constant. These checks guard against slips. The proofs do not use them.

The identity and the unique crossing for $s > 0$ in Proposition 3.2 are also checked in Lean 4. The asymptotic results, the product curve and the theorem on turns only are not. The data availability statement says where the scripts and the Lean files are.

Open questions. The first-order heuristic at the start of Section 3 applies the face selection of [3] to a linear function of the composition. We prove its conclusion for the planar walk, but not such a rule in general. The errors in Theorem 3.5 and Corollary 3.6 are only $o(1)$, and Theorem 3.5 needs $T \le N^{1/8}$. The next term of the crossing equation when turns dominate is open, and so is the crossover when $s - 2r$ tends to 0 as N grows. A heuristic suggests that the next term comes from replacing T by $T + \frac{1}{4r} - \frac{1}{2(r+s)}$ in Theorem 3.5. For $s = 0$ this agrees with the term $1/(8z_N)$ of the crossing equation in Section 2, through Theorem 3.9(c). Off the product curve of Proposition 3.7 we have no closed form for o_N or for the crossing, and Corollary 3.8 only brackets the crossing. With turns only, the computations give $N(v_N^* - u_N) = 0.37\,L^2/N$ at $N = 96$ and $0.38\,L^2/N$ at $N = 10240$. So the order L^2/N in Theorem 3.9(c) seems to be right, but we do not prove a lower bound of this order. The walk on $\mathbb{Z}^d$ with $d \ge 3$ is open as well.

Acknowledgments

I thank Peter Kagey for posing the problem. No funding was received for this work, and the author has no competing interests to declare.

Data availability statement

The code, data and Lean files that support this paper are openly available at `https://github.com/apemm/Kagey-Problems`, in the folder `kagey-problems/problem131` at the release `paperE-arxiv-v1`. The scripts are in the subfolder `paperE` and need numpy and scipy. To rerun the checks, run `python -B verify_all.py --paper e` in the folder `problem131`. The largest cases, such as the planar crossings for $N \ge 1280$, run only with the option `--full`. Section 4 describes what each check does. The Lean 4 proofs are in the subfolder `lean/PaperC`, whose file `MAP.md` matches each statement to its Lean theorems. The file `SUPPLEMENT.md` lists each check, the two methods behind it and the predictions that we wrote down before most of the computations.

A note on the use of AI

All mathematical claims, proofs and computations in this paper are my responsibility.

A Proofs for the planar walk

This appendix has three parts. We first prove the bound on $o_N^{(4)}$ in Proposition 3.2 and Theorem 3.3. Then we prove the local limit of Theorem 3.5, and finally parts (b) and (c) of Theorem 3.9. We identify $\mathbb{R}^2$ with $\mathbb{C}$, so $e_z = \mathrm{i}^z$ and $Y_N = \sum_{t=0}^{m} \ell_t \mathrm{i}^{Z_t}$ in the notation of Lemma 3.1. Let $\mathrm{bin}(m) = \binom{N-1}{m}(\beta h)^m p_0^{N-1-m}$ be the law of m, with mean $\bar{m} = (N-1)\beta h$. We write $\mathbb{P}_z$ and $\mathbb{E}_z$ for the skeleton walk of Lemma 3.1(c) started at z, $\mathbb{P}$ for its uniform start, and ξ for one of its increments. Put $\varkappa = s/\beta$, so $\varkappa < 1$ when $r > 0$. For a skeleton, let a, b, c, d be the numbers of runs going east, west, north and south. So $a + b + c + d = m + 1$, and the path uses all four directions exactly when $a, b, c, d \ge 1$. Throughout, $N = 2\nu$, and $[\,\cdot\,]$ is the indicator of the statement inside.

For $k \ge 1$ let $\chi_k(X) = \binom{X-1}{k-1}[X \ge 1]$, and let $\chi_0(X) = [X = 0]$. Then $\chi_k(X)$ counts the compositions of X into k positive parts.

Lemma A.1 (return counts)**.** *Let Z be a skeleton with m switches and counts a, b, c, d. The number of compositions ℓ of N into $m+1$ parts with $\sum_t \ell_t \mathrm{i}^{Z_t} = 0$ is*

$$\Omega_N(a,b,c,d) = \sum_{X=0}^{\nu} \chi_a(X)\chi_b(X)\chi_c(\nu - X)\chi_d(\nu - X).$$

If $a, b, c, d \geq 1$, then $m \geq 3$ and $\Omega_N(a,b,c,d) \leq \frac{1}{2}\binom{N}{m-2}$.

Proof. We first prove the formula. The sum vanishes exactly when the runs going east and the runs going west have the same total length X, and those going north and those going south both have total length $\nu - X$. Given X, the run lengths in the four directions can be chosen separately, in $\chi_a(X)$, $\chi_b(X)$, $\chi_c(\nu - X)$ and $\chi_d(\nu - X)$ ways. This gives the formula.

Now let $a, b, c, d \geq 1$, so that $m + 1 = a + b + c + d \geq 4$. If $m = 3$, there is one run in each direction and $\Omega_N = \nu - 1 < \frac{1}{2}\binom{N}{1}$. Let $m \geq 4$. We choose all but four of the run lengths freely and show that few choices remain for the last four. For each direction z pick one run t_z with $Z_{t_z} = z$, and call these four runs free. Suppose the other $m - 3$ run lengths are chosen and add up to $N - j$. There are $\binom{N-j-1}{m-4}$ such choices (the compositions of $N - j$ into $m - 3$ positive parts). The four free lengths then add up to j, and $Y_N = 0$ fixes the two differences $\ell_{t_0} - \ell_{t_2}$ and $\ell_{t_1} - \ell_{t_3}$. So the sum $\ell_{t_0} + \ell_{t_2}$ determines all four free lengths. This sum lies in $[2, j-2]$, because each free length is at least 1, and it has the parity of the fixed difference $\ell_{t_0} - \ell_{t_2}$. So at most $j/2$ choices of the free lengths remain, and

$$\Omega_N(a,b,c,d) \leq \sum_{j=0}^{N-1} \frac{j}{2}\binom{N-1-j}{m-4} = \frac{1}{2}\binom{N}{m-2},$$

where the identity counts the subsets of $\{0, \ldots, N-1\}$ of size $m - 2$ by their second smallest element j. □

By Lemma 3.1(b) and (d), $\varpi_N = \Omega_N(a,b,c,d)/\binom{N-1}{m}$ is the probability that $Y_N = 0$ given m and Z, so

$$o_N^{(4)} = \sum_{m \geq 3} \mathrm{bin}(m)\, \mathbb{E}\big[\varpi_N \mathbf{1}\{a,b,c,d \geq 1\} \,\big|\, m\big]. \tag{4}$$

Lemma A.2 (skeleton returns)**.** *Let $W_m = \sum_{t=0}^{m} \mathrm{i}^{Z_t} = (a-b) + \mathrm{i}(c-d)$.* (a) *For every start z, $\mathbb{E}_z|W_m|^2 \leq m + 1$.* (b) *If $s > 0$ and m is odd, then $\mathbb{P}(W_m = 0) \geq \varkappa/(36(m+1))$.* (c) *If $s = 0$ and $m \equiv 3 \pmod 4$, then $\mathbb{P}(W_m = 0) \geq 1/(m+1)$.*

Proof. (a) Since $\mathbb{E}\,\mathrm{i}^{\xi} = \frac{r}{\beta}(\mathrm{i} + \mathrm{i}^{-1}) + \frac{s}{\beta}\mathrm{i}^2 = -\varkappa$, we get $\mathbb{E}_z|W_m|^2 = \sum_{t,t'} \mathbb{E}_z \mathrm{i}^{Z_t - Z_{t'}} = m + 1 + 2\sum_{k=1}^{m}(m+1-k)(-\varkappa)^k$. The last sum alternates in sign, starts negative and has nonincreasing absolute terms, so it is at most 0.

(b) We cut the skeleton in the middle. If the switch in the middle is a reversal, then the second half is a copy of the first half, read backwards and rotated by a half turn. So the two halves cancel as soon as two independent copies of the same sum agree, and this has probability of order $1/m$ because the sum takes about m values.

Write $m = 2m_1 + 1$ and split $W_m = W' + W''$, where W' sums over $t \leq m_1$ and W'' over $t > m_1$. Each half has $m_1 + 1$ runs. For the first half, the law of ξ is symmetric and the uniform law is stationary, so the reversed sequence $(Z_{m_1}, \ldots, Z_0)$ is again the walk from a uniform start. Hence $\mathbb{P}(W' = y,\, Z_{m_1} = z) = \frac{1}{4}\mathbb{P}_z(W_{m_1} = y)$. For the second half, the switch after run m_1 is a

reversal with probability $\varkappa$, independently of $Z_0, \dots, Z_{m_1}$. In that case, given $Z_{m_1} = z$, the second half is the walk from $z+2$, so W'' has the law of W_{m_1} under $\mathbb{P}_{z+2}$. A half turn maps the walk from z to the walk from $z+2$ and changes the sign of the sum, so $\mathbb{P}_{z+2}(W_{m_1} = -y) = \mathbb{P}_z(W_{m_1} = y)$. Summing over z and over the value y of W',

$$\mathbb{P}(W_m = 0) \geq \frac{\varkappa}{4} \sum_z \sum_y \mathbb{P}_z(W_{m_1} = y)^2.$$

It remains to bound the inner sum from below. Let $R^2 = m+1$. By (a), $\mathbb{E}_z|W_{m_1}|^2 \leq m_1+1 = R^2/2$, so Chebyshev's inequality gives $\mathbb{P}_z(|W_{m_1}| \leq R) \geq \frac{1}{2}$. At most $9R^2$ Gaussian integers have modulus at most R. So by the Cauchy–Schwarz inequality the inner sum is at least $(\frac{1}{2})^2/(9R^2) = 1/(36R^2)$. Summing over the four values of z gives (b).

(c) Here we can compute the probability exactly, because the real and imaginary parts of W_m turn out to be independent sums of fair signs. Every switch is a turn, so $\mathrm{i}^{Z_t} = \mathrm{i}^{Z_{t-1}} \cdot \mathrm{i}\epsilon_t$ with independent fair signs ϵ_t. Hence $\mathrm{i}^{Z_t} = \mathrm{i}^{Z_0}\mathrm{i}^t\sigma_t$, where $\sigma_0 = 1$ and $\sigma_t = \epsilon_1 \cdots \epsilon_t$ are independent fair signs for $t \geq 1$. Write $m = 4k-1$. Then $\mathrm{i}^{-Z_0}W_m$ has real part $\sum_{t \text{ even}}(-1)^{t/2}\sigma_t$, which is 1 plus a sum of $2k-1$ fair signs, and an independent imaginary part that is a sum of $2k$ fair signs. So

$$\mathbb{P}(W_m = 0) = \binom{2k-1}{k}\binom{2k}{k}2^{1-4k} = \prod_{j=1}^{k} \frac{(2j-1)^2}{(2j)^2} \geq \frac{1}{4}\prod_{j=2}^{k} \frac{2j-2}{2j} = \frac{1}{4k} = \frac{1}{m+1}.$$

□

Lemma A.3 (balanced skeletons)**.** *If $a = b \geq 1$, $c = d \geq 1$ and $m = 2(a+c)-1 \leq N-1$, then $\Omega_N(a,b,c,d)/\binom{N-1}{m} \geq \sqrt{m+1}/(2N^2)$.*

Proof. We bound Ω_N from below by the Cauchy–Schwarz inequality. The ratio $\Omega_N/\binom{N-1}{m}$ then becomes a hypergeometric probability at its mode, and a mode of a law with standard deviation of order $\sqrt{k}$ has probability of order $1/\sqrt{k}$.

Put $k = a+c = (m+1)/2$, so $2 \leq k \leq \nu$. Only $1 \leq X \leq \nu-1$ contribute to Ω_N. The sum of $\binom{X-1}{a-1}\binom{\nu-X-1}{c-1}$ over these X is $\binom{\nu-1}{k-1}$, since both count the compositions of ν into k positive parts (sorted by the sum X of the first a parts). So the Cauchy–Schwarz inequality gives

$$\Omega_N = \sum_{X=1}^{\nu-1} \binom{X-1}{a-1}^2\binom{\nu-X-1}{c-1}^2 \geq \frac{1}{\nu}\binom{\nu-1}{k-1}^2 = \frac{k}{\nu^2}\binom{\nu-1}{k-1}\binom{\nu}{k}.$$

Next, $\binom{\nu-1}{k-1}\binom{\nu}{k}/\binom{2\nu-1}{2k-1} = \mathbb{P}(H = k-1)$, where H is the number of red balls among $2k-1$ balls drawn without replacement from $\nu-1$ red and ν blue ones. We show that $k-1$ is a mode of H. If $k = \nu$, then $H = k-1$ always. Otherwise the ratio $\mathbb{P}(H = j+1)/\mathbb{P}(H = j) = (\nu-1-j)(2k-1-j)/((j+1)(\nu-2k+2+j))$ decreases in j, is $\frac{\nu-k}{\nu-k+1} < 1$ at $j = k-1$ and exceeds 1 at $j = k-2$. So $k-1$ is a mode. Now $\operatorname{Var} H \leq (2k-1)/4$, so Chebyshev's inequality puts mass at least $\frac{3}{4}$ on the integers within $\sqrt{2k-1}$ of $\mathbb{E}H$. There are at most $3\sqrt{2k}$ of them, and the mode has at least the average mass among them, so $\mathbb{P}(H = k-1) \geq 1/(4\sqrt{2k})$. As $\binom{N-1}{m} = \binom{2\nu-1}{2k-1}$, we get $\Omega_N/\binom{N-1}{m} \geq k/(4\nu^2\sqrt{2k}) = \sqrt{m+1}/(2N^2)$. □

Proof of the bound on $o_N^{(4)}$ in Proposition 3.2. Note that $(\beta h)^m p_0^{N-1-m} = p_0^{N-1}w^m$. By (4) and Lemma A.1,

$$o_N^{(4)} \leq \frac{p_0^{N-1}}{2}\sum_{m=3}^{N-1}\binom{N}{m-2}w^m \leq \frac{p_0^{N-1}w^2}{2}\big((1+w)^N - 1\big) = \frac{w^2}{2p_0}\big(1-p_0^N\big),$$

since $1 + w = 1/p_0$. Dividing by c_N gives the bound on E_N. □

Proof of Theorem 3.3. We first prove three bounds on o_N/c_N, namely an upper bound, a lower bound from the axes and a lower bound from the paths with all four directions. Parts (a) to (c) then follow by inserting $h = \tau L/N$.

The upper bound. Since $I_1 \le I_0 \le e^z$ for $z \ge 0$, we have $\psi(z) \le 2ze^z$, and (2) gives $2S_N(u) \le \frac{4}{N}\psi(Nu) \le 8ue^{Nu}$. With Proposition 3.2 and $(1+w)^N \le e^{Nw}$,

$$\frac{o_N}{c_N} \le 8ue^{Nu} + 2w^2 e^{Nw}. \tag{5}$$

The first term bounds the axis part and the second the four-direction part.

The lower bound from the axes. Since $o_N \ge o_N^{(0)}$, (2) with $z = Nu$ gives

$$\frac{o_N}{c_N} \ge 2S_N(u) \ge \frac{4}{N}\Big(1 - \frac{z(z+2)}{2N}\Big)\psi(z). \tag{6}$$

The lower bound from four directions. We show that there are $c' > 0$ and T_0, depending only on r and s, with

$$o_N^{(4)} \ge \frac{c'}{N^2\sqrt{T}} \qquad \text{whenever } T = Nh \ge T_0 \text{ and } \beta h \le \tfrac{1}{2}. \tag{7}$$

We count only the balanced skeletons, namely those with $a = b \ge 1$ and $c = d \ge 1$. A skeleton with m switches is balanced with probability of order $1/m$ (Lemma A.2), and a balanced skeleton returns to the origin with probability at least $\sqrt{m+1}/(2N^2)$ (Lemma A.3). The product is of order $N^{-2}m^{-1/2}$, and m is typically close to $\bar m \approx \beta T$.

Lemma A.2 needs the right parity of m. So let $\mathcal{M}$ be the set of odd m if $s > 0$, and the set of $m \equiv 3 \pmod 4$ if $s = 0$, and let $m \in \mathcal{M}$. A skeleton is balanced exactly when $W_m = 0$ and it uses both axes. A skeleton that stays on one axis switches only by reversals, so these skeletons have probability at most $\varkappa^m$ given m, and there are none if $s = 0$. Since $\varkappa < 1$, Lemma A.2 gives m_2 and $c_2 > 0$ with

$$\mathbb{P}(a = b \ge 1,\ c = d \ge 1 \mid m) \ge \frac{c_2}{m+1} \qquad \text{for } m \in \mathcal{M},\ m \ge m_2$$

(take $c_2 = \varkappa/72$ if $s > 0$ and $c_2 = 1$ if $s = 0$). On this event Lemma A.3 gives $\varpi_N \ge \sqrt{m+1}/(2N^2)$. So in (4) each $m \in \mathcal{M}$ with $m \ge m_2$ contributes at least $\mathrm{bin}(m)\, c_2/(2N^2\sqrt{m+1})$. Let $J = [\bar m/2, 2\bar m]$ and suppose that $\bar m \ge 2m_2$. We keep only the terms with $m \in \mathcal{M} \cap J$, for which $m \ge m_2$ and $m+1 \le 2\bar m + 1$. This gives $o_N^{(4)} \ge c_2 \mathbb{P}(m \in \mathcal{M} \cap J)/(2N^2\sqrt{2\bar m + 1})$. It remains to show that $\mathbb{P}(m \in \mathcal{M} \cap J) \ge \frac{1}{8}$ when $\bar m$ is large. The Chernoff bounds for the binomial law are $\mathbb{P}(m \le \bar m/2) \le e^{-\bar m/8}$ and $\mathbb{P}(m \ge 2\bar m) \le e^{-\bar m/3}$, so $\mathbb{P}(m \notin J) \le 2e^{-\bar m/8}$. For the parity recall that $\beta h \le \frac{1}{2}$. If $s > 0$, then $\mathbb{P}(m \text{ odd}) = \frac{1}{2}(1 - (1-2\beta h)^{N-1}) \ge \frac{1}{2}(1 - e^{-2\bar m})$. If $s = 0$, the roots of unity filter gives

$$\mathbb{P}(m \equiv 3 \bmod 4) = \frac{1}{4}\sum_{k=0}^{3} \mathrm{i}^{-3k}\big(1 - \beta h + \beta h\, \mathrm{i}^k\big)^{N-1} \ge \frac{1}{4} - \frac{3}{4}e^{-\bar m/2},$$

since $|1 - \beta h + \beta h\, \mathrm{i}^{\pm 1}|^2 = 1 - 2\beta h(1-\beta h) \le e^{-\beta h}$ and $|1 - 2\beta h| \le e^{-2\beta h}$. In both cases $\mathbb{P}(m \in \mathcal{M}) \ge \frac{1}{4} - \frac{3}{4}e^{-\bar m/2}$, so $\mathbb{P}(m \in \mathcal{M} \cap J) \ge \frac{1}{4} - \frac{3}{4}e^{-\bar m/2} - 2e^{-\bar m/8} \ge \frac{1}{8}$ when $\bar m$ is large. As $\bar m \ge \beta T/2$, and $2\bar m + 1 \le 3\beta T$ when $\beta T \ge 1$, this proves (7) with $c' = c_2/(16\sqrt{3\beta})$.

(a) Let $h = \tau L/N$ and $T = \tau L$. Then $u, w = O(L/N)$, $Nu = s\tau L + O(L^2/N)$ and $Nw = \beta\tau L + O(L^2/N)$. So $e^{Nu} = O(N^{s\tau})$ and $e^{Nw} = O(N^{\beta\tau})$, and (5) gives $o_N/c_N = O(LN^{s\tau-1} + L^2N^{\beta\tau-2})$. If $\tau < \tau^*$, then $s\tau < 1$ and $\beta\tau < 2$, and this is at most $N^{-\gamma}$ for large N whenever $0 < \gamma < \min(1 - s\tau, 2 - \beta\tau)$.

Now let $\tau > \tau^*$, so $s\tau > 1$ or $\beta\tau > 2$. If $s\tau > 1$, we use the axes. Then $z = Nu \to \infty$ and $z^2/N \to 0$. As $I_0(z) + I_1(z) \sim 2e^z/\sqrt{2\pi z}$ [24, Eq. 10.40.1], we have $\psi(z) \sim \sqrt{2z/\pi}\, e^z$, and $e^z = N^{s\tau}(1 + o(1))$. So (6) gives $o_N/c_N \geq N^{s\tau-1}$ for large N. If $\beta\tau > 2$, we use the four directions. Then (7) and $c_N^{-1} = 4p_0^{-(N-1)} \geq 4e^{\beta T - \beta h}$ give $o_N/c_N \geq 4c'e^{-\beta h}N^{\beta\tau-2}(\tau L)^{-1/2}$, which is at least $N^{(\beta\tau-2)/2}$ for large N.

(b) Let $\tau < \tau^* < \tau'$. By (a), for large N the ratio o_N/c_N is below 1 at $h = \tau L/N$ and above 1 at $h = \tau' L/N < 1/\beta$. It increases in h by Proposition 3.2, so $\tau L < T_N^* < \tau' L$. As τ and τ' are arbitrary, this is (b).

(c) We evaluate the two parts of the identity $2S_N(u_N^*) + E_N = 1$, which holds at the crossing by Proposition 3.2, with the bounds of (5). Put $\tau_N = T_N^*/L$, so $\tau_N \to \tau^*$ by (b), and $w^* = \beta h_N^*/(1 - \beta h_N^*)$. As in (a), $u_N^*, w^* = O(L/N)$, $Nu_N^* = s\tau_N L + o(1)$ and $Nw^* = \beta\tau_N L + o(1)$. If $s > 2r$, then $\tau^* = 1/s$ and $\beta\tau^* < 2$. So $E_N \leq 2w^{*2}e^{Nw^*} = N^{\beta/s-2+o(1)} \to 0$, and $2S_N(u_N^*) \to 1$. The two axes carry the same mass by symmetry. If $s < 2r$, then $\tau^* = 2/\beta$ and $s\tau^* < 1$. If $s = 0$, then $u_N^* = 0$ and $S_N(u_N^*) = 0$. If $s > 0$, then $2S_N(u_N^*) \leq 8u_N^* e^{Nu_N^*} = N^{2s/\beta-1+o(1)}$, which is at most $N^{-\gamma}$ for large N whenever $0 < \gamma < 1 - 2s/\beta$. □

A.1 The four-direction local limit

Idea of the proof. Given the skeleton, the run lengths are close to the gaps between m uniform points in $[0, N]$. So the total lengths in the four directions are close to N times a Dirichlet vector (y_0, y_1, y_2, y_3) with parameters (a, c, b, d). Then $N^2\varpi_N$ should be close to twice the density of $(y_0 - y_2, y_1 - y_3)$ at 0, where the factor 2 is the parity of Y_N. This is the quantity $g(a, b, c, d)$ below. Divided by m, it is close to a Gaussian function of $W_m/\sqrt{m}$, and W_m satisfies a central limit theorem. Steps 1 and 2 below remove the skeletons with too few or too many switches and those that use some direction rarely. Step 3 compares $N^2\varpi_N$ with g, Step 4 averages g over the skeleton, and Step 5 puts the steps together.

Proof of Theorem 3.5. We use the notation of the start of this appendix. Let $A = a + b$ and $B = c + d$, so $A + B = m + 1$, and put $\Gamma_4 = (a-1)!\,(b-1)!\,(c-1)!\,(d-1)!$ and

$$g(a, b, c, d) = \frac{m(m-1)}{2} \cdot \frac{\binom{A-2}{a-1}}{2^{A-2}} \cdot \frac{\binom{B-2}{c-1}}{2^{B-2}} = \frac{m!\,(A-2)!\,(B-2)!}{2^{m-2}\,(m-2)!\,\Gamma_4}.$$

Let $J_N = [\bar{m}/2, 2\bar{m}]$, $\mathcal{G}_m = \{a, b, c, d \geq m/8\}$ and $m_0 = 20$. Note that $\bar{m} = \beta T(1 - 1/N) \to \infty$ and $\bar{m} \leq \beta N^{1/8}$. Below, C is a constant that depends only on r and s and may change from line to line.

Step 1 (few or many switches). We show that the terms of (4) with $m \notin J_N$ contribute $o(1)$ to $N^2 o_N^{(4)}$. For this we need a bound on ϖ_N that holds for every skeleton. If $a, b, c, d \geq 1$ and $m \leq N/5$, then Lemma A.1 and $\binom{N}{m-2}/\binom{N-1}{m} = Nm(m-1)/((N-m+2)(N-m+1)(N-m))$ give $\varpi_N \leq m^2/N^2$. If $m > N/5$, then $\varpi_N \leq 1 < 25m^2/N^2$. So $N^2\varpi_N \leq 25m^2$ whenever the path uses all four directions. As in the proof of Theorem 3.3, the Chernoff bounds give $\mathbb{P}(m \notin J_N) \leq 2e^{-\bar{m}/8}$. Also $\mathbb{E}m^4 = \sum_{j=1}^4 \left\{{4 \atop j}\right\}(N-1)_j(\beta h)^j \leq 15\bar{m}^4$, where $\left\{{4 \atop j}\right\} = 1, 7, 6, 1$ are Stirling numbers of the second kind, $(N-1)_j$ is a falling factorial, and $\bar{m} \geq 1$. By the Cauchy–Schwarz inequality, the

terms of (4) with $m \notin J_N$ contribute at most $25\,\mathbb{E}[m^2\mathbf{1}\{m \notin J_N\}] \le 25\sqrt{30}\,\bar{m}^2 e^{-\bar{m}/16} = o(1)$ to $N^2 o_N^{(4)}$.

Step 2 (unbalanced skeletons). We show that the skeletons outside $\mathcal{G}_m$ contribute $O(1)$ to $N^2 o_N^{(4)}$. The reason is that each of a, b, c, d is close to $\frac{m+1}{4}$, with a deviation of order $\sqrt{m}$. To see this we write the four counts through three sums that are nearly martingales. For $k = 1, 2, 3$ let $V_k = \sum_{t=0}^{m} \mathrm{i}^{kZ_t}$. The number of t with $Z_t = z$ is $\frac{m+1}{4} + \frac{1}{4}\sum_{k=1}^{3} \mathrm{i}^{-kz} V_k$. Let $\vartheta_k = \mathbb{E}\,\mathrm{i}^{k\xi}$, so $\vartheta_1 = \vartheta_3 = -\varkappa$ and $\vartheta_2 = 2\varkappa - 1$. Then $\mathbb{E}[\mathrm{i}^{kZ_t} \mid Z_0, \dots, Z_{t-1}] = \vartheta_k \mathrm{i}^{kZ_{t-1}}$, so $M_m^{(k)} = \sum_{t=1}^{m} (\mathrm{i}^{kZ_t} - \vartheta_k \mathrm{i}^{kZ_{t-1}})$ is a martingale with increments of modulus at most 2, and

$$(1 - \vartheta_k) V_k = M_m^{(k)} + \mathrm{i}^{kZ_0} - \vartheta_k \mathrm{i}^{kZ_m}.$$

Here $1 - \vartheta_k \ge \min(1, 4r/\beta) > 0$. Burkholder's inequality [23, Theorem 2.10], applied to the real and imaginary parts, bounds $\mathbb{E}|M_m^{(k)}|^4$ by a constant times the second moment of the sum of the m squared increments. This sum is at most $4m$, so $\mathbb{E}|M_m^{(k)}|^4 \le Cm^2$. By the last display the same bound holds for V_k, namely $\mathbb{E}|V_k|^4 \le Cm^2$. Hence $\mathbb{E}(a - \frac{m+1}{4})^4 \le Cm^2$, and the same holds for b, c, d. Since $\frac{m+1}{4} - \frac{m}{8} \ge \frac{m}{8}$, Markov's inequality gives $\mathbb{P}(\mathcal{G}_m^c \mid m) \le C/m^2$. Every skeleton in $\mathcal{G}_m$ uses all four directions. By Step 1, the skeletons with all four directions that are not in $\mathcal{G}_m$, with $m \in J_N$, contribute at most $\sum_m \mathrm{bin}(m)\, 25m^2\, C/m^2 \le 25C$ to $N^2 o_N^{(4)}$.

Step 3 (run lengths given the skeleton). We show that $N^2 \varpi_N = g(a, b, c, d)(1 + o(1))$ uniformly over the skeletons in $\mathcal{G}_m$ with $m \in J_N$. The reason is that each factor $\chi_a(X) = \binom{X-1}{a-1}$ of Ω_N is close to $X^{a-1}/(a-1)!$ when X is much larger than a^2. So Ω_N is close to a sum of $X^{A-2}(\nu - X)^{B-2}$ over X, and this sum is close to a Beta integral. For large N the $m \in J_N$ satisfy $m_0 \le m \le 2\beta N^{1/8} \le N^{1/6}$. On $\mathcal{G}_m$ we have $A, B \ge m/4$, so $A - 1, B - 1 \ge m/8$. Put $f(X) = X^{A-2}(\nu - X)^{B-2}$. For $X \ge 1$ we have $\chi_a(X) = \frac{X^{a-1}}{(a-1)!} \prod_{1 \le j < a} (1 - j/X)_+$, where $(y)_+ = \max(y, 0)$, and similarly for b, c, d. The terms $X = 0$ and $X = \nu$ of Ω_N vanish. So

$$\Omega_N(a, b, c, d) = \frac{1}{\Gamma_4} \sum_{X=1}^{\nu-1} f(X)\omega(X),$$

where $\omega(X) \in [0, 1]$ is the product of the four products. The function f vanishes at 0 and ν and is unimodal, and its mode $X^* = \nu(A - 2)/(m - 3)$ lies in $[\nu/6, 5\nu/6]$ since $A, B \ge m/4$ and $m \ge 18$. On each monotone piece of f every value $f(X)$ lies between the integrals of f over the two unit intervals next to X. So $|\sum_{j \le X \le k} f(X) - \int_j^k f| \le 2 \max f$ for all integers $0 \le j \le k \le \nu$. Also, $f / \int_0^\nu f$ is the density of ν times a Beta variable with parameters $A - 1$ and $B - 1$, and $\int_0^\nu f = \nu^{m-2} \mathrm{B}(A - 1, B - 1)$.

We need three estimates. The first says that the ends of the integral are small, the second that ω is close to 1 away from the ends, and the third that $\max f$ is small compared with the integral. First, this Beta law is the law of the $(A - 1)$-th smallest of $m - 2$ independent uniform points in $[0, 1]$. A union bound over sets of $A - 1$ points gives, for $0 < t < 1$,

$$\mathbb{P}(\mathrm{Beta} \le t) \le \binom{m-2}{A-1} t^{A-1} \le \Big(\frac{e(m-2)t}{A-1}\Big)^{A-1} \le (8et)^{A-1}.$$

We take $t = 2\nu^{-1/4}$. Since $A - 1 \ge 2$ and $16e\nu^{-1/4} < 1$ for large N, this gives $\int_0^{\nu^{3/4}+1} f \le (16e)^2 \nu^{-1/2} \int_0^\nu f$, and the same bound holds near ν. Second, for $\nu^{3/4} \le X \le \nu - \nu^{3/4}$, the inequality $1 - \prod_j (1 - t_j)_+ \le \sum_j t_j$ for $t_j \ge 0$ gives $1 - \omega(X) \le (a^2 + b^2 + c^2 + d^2)/(2\nu^{3/4}) \le m^2 \nu^{-3/4}$. Third, on $[\nu/12, 11\nu/12]$ we have $(\log f)'' = -(A - 2)/X^2 - (B - 2)/(\nu - X)^2 \ge -144m/\nu^2$. This

interval contains every X with $|X - X^*| \le \nu/(12\sqrt{m})$, and for these X Taylor's formula at X^* gives $f(X) \ge e^{-1/2} \max f$. So $\int_0^\nu f \ge e^{-1/2} \max f \cdot \nu/(6\sqrt{m})$, that is, $\max f \le 6e^{1/2}\sqrt{m}\,\nu^{-1} \int_0^\nu f$.

Now we put the three estimates together. For the upper bound, $\sum_{X=1}^{\nu-1} f\omega \le \sum_X f \le \int_0^\nu f + 2\max f$. For the lower bound we sum only over $\nu^{3/4} \le X \le \nu - \nu^{3/4}$, use the second estimate for ω, compare the sum of f with its integral, and use the first estimate for the two ends of the integral. This gives

$$\sum_{X=1}^{\nu-1} f\omega \ge \big(1 - m^2\nu^{-3/4}\big)\Big(\int_0^\nu f - 2(16e)^2\nu^{-1/2}\int_0^\nu f - 2\max f\Big).$$

Since $m^2 \le N^{1/3}$ and $\sqrt{m}/\nu \le N^{-5/6}$, we get $\Gamma_4\Omega_N = \nu^{m-2}\mathrm{B}(A-1,B-1)(1+o(1))$, uniformly. Also $\binom{N-1}{m} = \frac{N^m}{m!}\prod_{j=1}^m(1-\frac{j}{N}) = \frac{N^m}{m!}(1+O(m^2/N))$. With $\nu^{m-2}/N^{m-2} = 2^{-(m-2)}$ and $\mathrm{B}(A-1,B-1) = (A-2)!\,(B-2)!/(m-2)!$ this gives $N^2\varpi_N = g(a,b,c,d)(1+o(1))$.

Step 4 (average over the skeleton). We show that $\mathbb{E}[g\,\mathbf{1}_{\mathcal{G}_m} \mid m]/m \to c_* = \frac{2(r+s)}{\pi\beta}$ as $m \to \infty$. This has three parts. We write g/m as a Gaussian function Ψ of the skeleton, we prove a central limit theorem for W_m, and we take the limit of the expectation of Ψ. For the first part we need a local limit for the fair binomial law. Let $0 \le k \le n$ and $y = (2k-n)/n$. If $|2k-n| \le n^{3/5}$, Stirling's formula gives

$$\begin{aligned}\sqrt{n}\binom{n}{k}2^{-n} &= \sqrt{\frac{2}{\pi(1-y^2)}}\,e^{-\frac{n}{2}[(1+y)\log(1+y)+(1-y)\log(1-y)]}\big(1+O(\tfrac{1}{n})\big)\\ &= \sqrt{\frac{2}{\pi}}\,e^{-\frac{(2k-n)^2}{2n}}\big(1+O(n^{-3/5})\big),\end{aligned}$$

since the bracket is $y^2 + O(y^4)$ and $ny^4 \le n^{-3/5}$. Both sides decrease as $|2k-n|$ grows, so outside this range both are $O(e^{-n^{1/5}/3})$. Hence

$$\sup_{0\le k\le n}\Big|\sqrt{n}\binom{n}{k}2^{-n} - \sqrt{\tfrac{2}{\pi}}\,e^{-(2k-n)^2/(2n)}\Big| = O(n^{-1/2}). \tag{8}$$

On $\mathcal{G}_m$ we have $A-2, B-2 \ge m/8$. We apply (8) with $n = A-2$, $k = a-1$ and with $n = B-2$, $k = c-1$, and note that $2(a-1) - (A-2) = a-b$. Then, uniformly on $\mathcal{G}_m$,

$$\frac{g}{m} = \frac{m-1}{\pi\sqrt{(A-2)(B-2)}}\exp\Big(-\frac{(a-b)^2}{2(A-2)} - \frac{(c-d)^2}{2(B-2)}\Big) + O(m^{-1/2}).$$

Now we replace $m-1$ by m, then $A-2$ by A and $B-2$ by B, and finally B by $m(1-\theta)$, where $\theta = A/m$. The last step is needed because $B = m(1-\theta)+1$. Each step changes the prefactor by a factor $1+O(1/m)$ and each exponential by $O(1/m)$, since $0 \le e^{-y/A_2} - e^{-y/A_1} \le \frac{A_2-A_1}{eA_1}$ for $y \ge 0$ and $0 < A_1 \le A_2$. So, uniformly on $\mathcal{G}_m$,

$$\frac{g}{m} = \Psi\Big(\frac{a-b}{\sqrt{m}}, \frac{c-d}{\sqrt{m}}, \frac{A}{m}\Big) + O(m^{-1/2}), \qquad \Psi(x_1,x_2,\theta) = \frac{e^{-x_1^2/(2\theta) - x_2^2/(2(1-\theta))}}{\pi\sqrt{\theta(1-\theta)}}.$$

On $\mathcal{G}_m$ we have $\theta \in [\frac14, \frac34 + \frac1m] \subset [\frac15, \frac45]$, as $m \ge 20$. Let $\tilde\Psi$ be Ψ with θ replaced by the nearest point of $[\frac15, \frac45]$. It is bounded and continuous on $\mathbb{R}^3$ and equals Ψ on $\mathcal{G}_m$.

Next we prove a central limit theorem for $W_m = (a-b) + \mathrm{i}(c-d)$. Put $\zeta_j = \mathrm{i}^{\xi_j} + \varkappa$, where $\xi_j = Z_j - Z_{j-1}$. The ζ_j are independent with $\mathbb{E}\zeta_j = 0$, $\mathbb{E}|\zeta_j|^2 = 1-\varkappa^2$ and $\mathbb{E}\zeta_j^2 = \vartheta_2 - \varkappa^2 = -(1-\varkappa)^2$. Since $\mathrm{i}^{Z_j} = \mathrm{i}^{Z_{j-1}}(\zeta_j - \varkappa)$,

$$(1+\varkappa)W_m = M_m + \mathrm{i}^{Z_0} + \varkappa\,\mathrm{i}^{Z_m}, \qquad M_m = \sum_{j=1}^{m} \mathrm{i}^{Z_{j-1}}\zeta_j,$$

and M_m is a martingale with increments of modulus at most 2. We take the skeletons for all m to be the first steps of one infinite walk, so the σ-fields are nested. Fix $\varphi \in \mathbb{R}$. For a constant υ with $|\upsilon| = 1$ we have $\mathbb{E}(\operatorname{Re}\upsilon\zeta_j)^2 = \frac{1}{2}(\mathbb{E}|\zeta_j|^2 + \operatorname{Re}\upsilon^2\mathbb{E}\zeta_j^2)$. We apply this with $\upsilon = e^{-\mathrm{i}\varphi}\mathrm{i}^{Z_{j-1}}$, for which $\upsilon^2 = e^{-2\mathrm{i}\varphi}(-1)^{Z_{j-1}}$. So the j-th increment of $\operatorname{Re}(e^{-\mathrm{i}\varphi}M_m)$ has conditional variance $\frac{1}{2}[(1-\varkappa^2) - (1-\varkappa)^2\cos(2\varphi)(-1)^{Z_{j-1}}]$. The average of these variances tends to $\frac{1}{2}(1-\varkappa^2)$ in probability, because $\sum_{t=0}^{m}(-1)^{Z_t} = V_2$ and $\mathbb{E}|V_2/m|^4 \le C/m^2$ by Step 2. The increments are bounded, so the martingale central limit theorem [23, Cor. 3.1] and the Cramér–Wold device give $M_m/\sqrt{m} \Rightarrow \mathcal{N}(0, \frac{1}{2}(1-\varkappa^2)I_2)$. The other two terms in the identity for $(1+\varkappa)W_m$ are bounded, so they do not matter after division by $\sqrt{m}$. Hence

$$\frac{W_m}{\sqrt{m}} \Rightarrow G \sim \mathcal{N}(0, \sigma_*^2 I_2), \qquad \sigma_*^2 = \frac{1-\varkappa}{2(1+\varkappa)} = \frac{r}{2(r+s)}.$$

Also $A = \frac{m+1}{2} + \frac{1}{2}V_2$, so $A/m \to \frac{1}{2}$ in probability, and $(W_m/\sqrt{m}, A/m) \Rightarrow (G, \frac{1}{2})$.

Now we take the limit. By Step 2, $\mathbb{E}[\tilde{\Psi}\,\mathbf{1}_{\mathcal{G}_m^c} \mid m] \le C/m^2$. So, since $\tilde{\Psi}$ is bounded and continuous,

$$\frac{\mathbb{E}[g\,\mathbf{1}_{\mathcal{G}_m} \mid m]}{m} = \mathbb{E}\,\tilde{\Psi}\Big(\frac{W_m}{\sqrt{m}}, \frac{A}{m}\Big) + O(m^{-1/2}) \to \mathbb{E}\,\Psi\big(G, \tfrac{1}{2}\big) = \frac{2}{\pi}\,\mathbb{E}e^{-|G|^2} = \frac{2}{\pi(1+2\sigma_*^2)} = c_*.$$

Step 5 (conclusion). By Steps 1 and 2, $N^2 o_N^{(4)} = \sum_{m\in J_N} \operatorname{bin}(m)\,\mathbb{E}[N^2\varpi_N\mathbf{1}_{\mathcal{G}_m} \mid m] + O(1)$. By Step 3 the expectation is $\mathbb{E}[g\,\mathbf{1}_{\mathcal{G}_m} \mid m](1+o(1))$, with $o(1)$ uniform in $m \in J_N$. Since $\min J_N \to \infty$, Step 4 turns this into $N^2 o_N^{(4)} = c_*(1+o(1))\sum_{m\in J_N}\operatorname{bin}(m)\,m + O(1)$. As in Step 1, $\mathbb{E}[m\mathbf{1}\{m \notin J_N\}] \to 0$, and $\bar{m} = \beta T(1+O(1/N))$. Hence $N^2 o_N^{(4)} = c_*\beta T(1+o(1)) + O(1) = \frac{2(r+s)}{\pi}T(1+o(1))$. □

A.2 Turns only

We prove parts (b) and (c) of Theorem 3.9. Throughout, a word is a word of length n in the letters $\pm$, its weight is x^j if it has j switches, and we say that it switches at step t if its letters t and $t+1$ differ ($1 \le t \le n-1$). The words of length n have total weight $2(1+x)^{n-1}$, and those with k letters + have total weight $S_{k,n-k}(x)$ if $1 \le k \le n-1$ and 1 if $k \in \{0, n\}$. We need one fact about Bessel functions. Let $F(y) = e^{-y}(I_0(y) + I_1(y))$ for $y > 0$, so that $\psi(y) = ye^yF(y)$. The integral forms $I_n(y) = \frac{1}{\pi}\int_0^\pi e^{y\cos\theta}\cos(n\theta)\,d\theta$ of I_0 and I_1 [24, Eq. 10.32.3] and the substitution $\varsigma = 1-\cos\theta$ give

$$F(y) = \frac{1}{\pi}\int_0^\pi e^{-y(1-\cos\theta)}(1+\cos\theta)\,d\theta = \frac{1}{\pi}\int_0^2 e^{-y\varsigma}\sqrt{\frac{2-\varsigma}{\varsigma}}\,d\varsigma.$$

So F is positive and decreasing. Replacing y by $y/2$ and ς by 2ς gives

$$F(y/2) = \frac{2}{\pi}\int_0^1 e^{-y\varsigma}\sqrt{\frac{1-\varsigma}{\varsigma}}\,d\varsigma.$$

This is at most $2F(y)$, because $1-\varsigma \le 2-\varsigma$ and the part of the integral for $F(y)$ over $[1,2]$ is nonnegative. Hence

$$F(y') \le 2F(y) \qquad \text{whenever } y/2 \le y' \le y. \tag{9}$$

Lemma A.4 (one switch of a balanced word)**.** *Let $N \ge 4$ be even, let $x > 0$, and put $z = Nx$ and $\eta = z(z+2)/(2N)$. For $1 \le t \le N-1$ let $S_N^{(t)}(x)$ be the total weight of the balanced words of length N that switch at step t. If $\eta \le \frac{1}{2}$, then $S_N^{(t)}(x) \le 2x + 8x\,S_N(x)$ for every t, and $\sum_t S_N^{(t)}(x) = xS_N'(x) \le (2z+2)S_N(x)$.*

In words, when $S_N(x)$ is of order one, as it is near the crossing, a balanced word switches at a given step with probability of order x, as a word without any condition does. A balanced word has at most $2z+2$ switches on average.

Proof. We cut a balanced word at the step t where it switches and bound the total weight of each piece. Reversing a word maps the balanced words that switch at t to those that switch at $N-t$, and it keeps the weight. So we may take $t \ge N/2$. A balanced word that switches at t is a word w_1 of length t followed by a word w_2 of length $N-t$ whose first letter differs from the last letter of w_1, and its weight is x times the product of the two weights. If w_1 has k letters $+$, then w_2 has $N/2-k$ of them, so $k \le N/2 \le t$ and $t-k \le N/2 \le t$.

Consider first the balanced words with $1 \le k \le t-1$. Fix w_2. Then k is fixed, and the words w_1 that can precede w_2 have total weight at most $S_{k,t-k}(x)$. By [2, Proposition 4.1] the bin polynomials grow coefficientwise as the bin moves toward the middle, that is, $S_{a,b} \le S_{a+1,b-1}$ coefficientwise when $a < b-1$. Since also $S_{a,b} = S_{b,a}$, this gives $S_{k,t-k}(x) \le S_t(x)$. The words w_2 of length $N-t$ have total weight at most $2(1+x)^{N-t-1} \le 2e^{(N-t)x}$. By (2) with $t \ge 2$ steps, $S_t(x) \le \frac{2}{t}\psi(tx) = 2xe^{tx}F(tx)$. So these words contribute at most

$$x \cdot 2xe^{tx}F(tx) \cdot 2e^{(N-t)x} = 4x^2e^zF(tx) \le 8x^2e^zF(z) = 4x \cdot \frac{2}{N}\psi(z) \le \frac{4x}{1-\eta}\,S_N(x) \le 8x\,S_N(x),$$

to $S_N^{(t)}(x)$. Here the first inequality is (9) with $z/2 \le tx \le z$, and the next one is the lower bound in (2). If $k = 0$ or $k = t$, then w_1 is constant and $t = N/2$, and w_2 is the constant word of the other letter. These two words have weight x each. This proves the first claim.

Each balanced word with j switches is counted j times in $\sum_t S_N^{(t)}$, so the sum is $xS_N'(x)$. The function $y \mapsto \log S_N(e^y)$ is convex, since S_N has nonnegative coefficients. So its derivative at $y = \log x$ is at most its increase over $[\log x, \log x + 1]$, that is, $xS_N'(x)/S_N(x) \le \log(S_N(ex)/S_N(x))$. By (2), $S_N(ex) \le \frac{2}{N}\psi(ez)$ and $S_N(x) \ge \frac{1}{2} \cdot \frac{2}{N}\psi(z)$, and $\psi(ez)/\psi(z) = e \cdot e^{(e-1)z}F(ez)/F(z) \le e^{1+(e-1)z}$. Hence $xS_N'(x)/S_N(x) \le 1 + \log 2 + (e-1)z \le 2z+2$. □

Proof of Theorem 3.9(b). A pair of balanced words with a common switch has a common switch at some step t, and the pairs that both switch at t have total weight $S_N^{(t)}(x)^2$. So $\Delta_N(x) \le \sum_t S_N^{(t)}(x)^2 \le \max_t S_N^{(t)}(x) \cdot \sum_t S_N^{(t)}(x)$, and Lemma A.4 gives the bound, because $z(z+2) \le N$ means $\eta \le \frac{1}{2}$. □

Proof of Theorem 3.9(c). We compare v_N^* with a value x_1 of the odds slightly above u_N, at which the bound of part (b) is already smaller than the gain in S_N^2.

Put $z_0 = Nu_N$. Since $S_N(u) \ge 2u$, we have $u_N \le \frac{1}{2}$ and $p_N \ge \frac{2}{3}$. With $z_0 = z_N/p_N$ and $z_N \le L$ [2, Proposition 3.8] this gives $z_0 \le \frac{3}{2}L$. By (2) and $S_N(u_N) = 1$, $\psi(z_0) \ge N/2$. Let $\mu = 40(z_0+2)^2/N$, which tends to 0, and let $x_1 > u_N$ be the root of $S_N(x_1) = 1+\mu$, with $z_1 = Nx_1$ and $\eta_1 = z_1(z_1+2)/(2N)$.

We first show that $z_1 \le z_0 + 1$ for large N. Since $(\log\psi)' \ge 1$, we have $\psi(z_0+1) \ge e\,\psi(z_0) \ge eN/2$. So by (2), at $x = (z_0+1)/N$ we get $S_N(x) \ge (1 - O(L^2/N))\,e$, which exceeds $1+\mu$ for large N. As S_N is increasing, $z_1 \le z_0 + 1$. In particular $z_1(z_1+2) \le N$ and $\mu \le 1$ for large N, so part (b) gives

$$\Delta_N(x_1) \le \big(2x_1 + 16x_1\big)(2z_1+2)\cdot 2 = \frac{72\,z_1(z_1+1)}{N} \le \frac{72(z_0+1)(z_0+2)}{N} \le 2\mu.$$

By part (a), $o_N/c_N = (1+\mu)^2 - \Delta_N(x_1) \ge 1$ at $v = x_1$. The ratio o_N/c_N increases with v, so $v_N^* \le x_1$. Now $\psi(z_1) \le \frac{N}{2}(1+\mu)/(1-\eta_1)$ by (2), and $(\log\psi)' \ge 1$. Hence

$$0 \le Nv_N^* - z_0 \le z_1 - z_0 \le \log\psi(z_1) - \log\psi(z_0) \le \log\frac{1+\mu}{1-\eta_1} = O\Big(\frac{L^2}{N}\Big),$$

where the first inequality is part (a). This is the first claim. For the second, $v = rh/(1-2rh)$ gives $rT_N^* = Nv_N^*/(1+2v_N^*)$, and $z_N = z_0/(1+u_N)$. Both v_N^* and u_N are $O(L/N)$, so $rT_N^* = Nv_N^* + O(L^2/N)$ and $z_N = z_0 + O(L^2/N)$. □

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